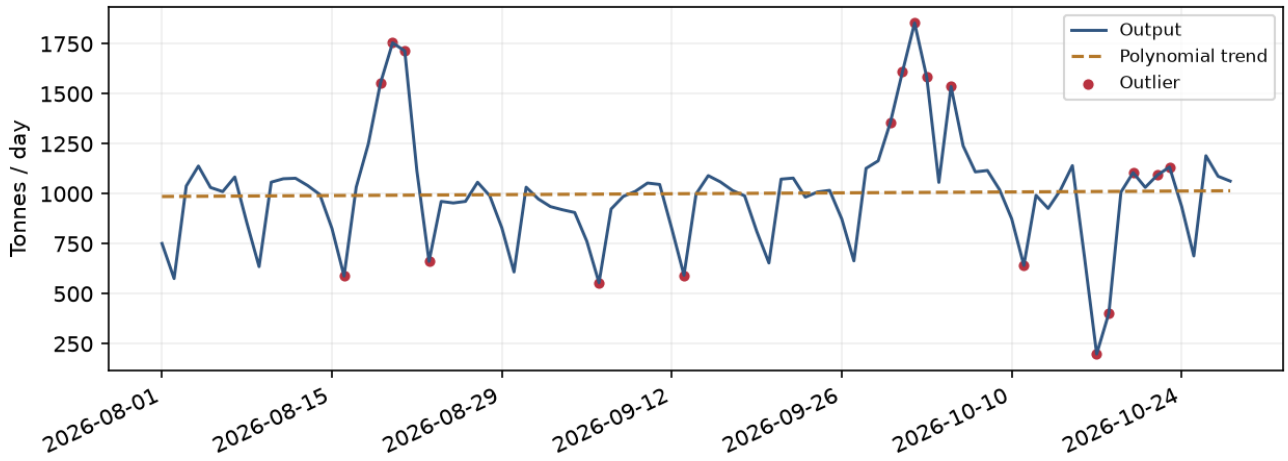
Mine production report

2026-08-01 to 2026-10-28

IQR factor: 1.5; z-score threshold: 3.0; previous observations: 7; moving gap: 35.0%; Grubbs alpha: 0.05; polynomial degree: 1.

Mine 01

Metric	Value
Observations	89
Total	89,035.70
Mean	1,000.40
Standard Deviation	273.19
Median	1,016.13
Iqr	212.84



Detected anomalous observations: 18

2026-08-16: drop

Output: 588.42 tonnes. Tests: Moving average.

2026-08-19: spike

Output: 1,553.59 tonnes. Tests: IQR, Moving average.

2026-08-20: spike

Output: 1,754.60 tonnes. Tests: IQR, Moving average.

2026-08-21: spike

Output: 1,713.92 tonnes. Tests: IQR, Moving average.

2026-08-23: drop

Output: 661.65 tonnes. Tests: Moving average.

2026-09-06: drop

Output: 554.87 tonnes. Tests: IQR, Moving average.

2026-09-13: drop

Output: 592.26 tonnes. Tests: Moving average.

2026-09-30: spike

Output: 1,355.57 tonnes. Tests: Moving average.

2026-10-01: spike

Output: 1,609.32 tonnes. Tests: IQR, Moving average.

2026-10-02: spike

Output: 1,853.80 tonnes. Tests: IQR, z-score, Moving average.

2026-10-03: spike

Output: 1,582.43 tonnes. Tests: IQR.

2026-10-05: spike

Output: 1,535.60 tonnes. Tests: IQR.

2026-10-11: drop

Output: 641.31 tonnes. Tests: Moving average.

2026-10-17: drop

Output: 198.37 tonnes. Tests: IQR, Moving average.

2026-10-18: drop

Output: 402.27 tonnes. Tests: IQR, Moving average.

2026-10-20: spike

Output: 1,105.22 tonnes. Tests: Moving average.

2026-10-22: spike

Output: 1,092.92 tonnes. Tests: Moving average.

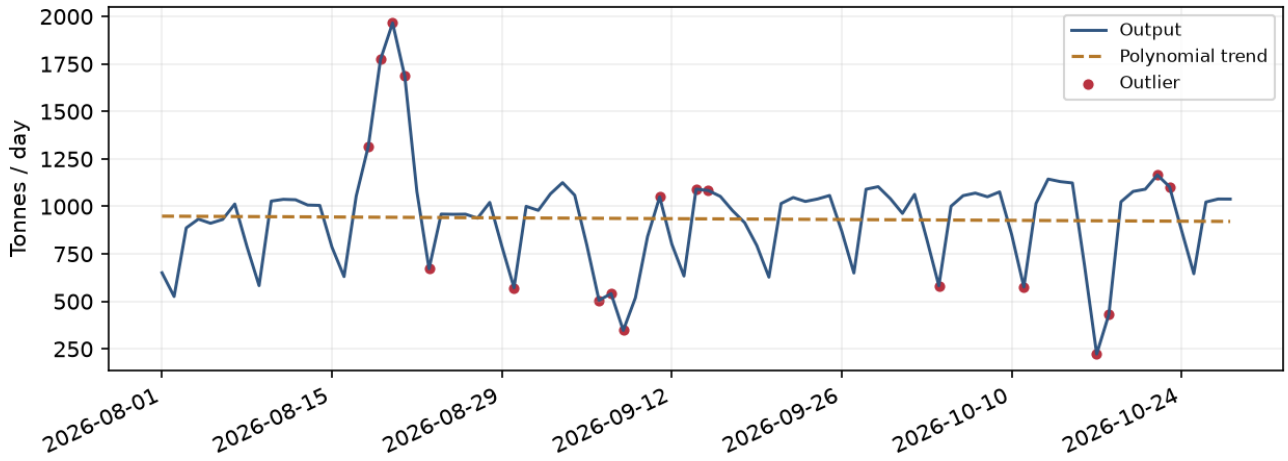
2026-10-23: spike

Output: 1,133.50 tonnes. Tests: Moving average.

Previous N observed rows; current point excluded; no test before a full window is available.
Iterative two-sided Grubbs test; assumes an approximately normal baseline.

Mine 02

Metric	Value
Observations	89
Total	83,208.08
Mean	934.92
Standard Deviation	266.39
Median	1,005.46
Iqr	263.30



Detected anomalous observations: 18

2026-08-18: spike

Output: 1,316.92 tonnes. Tests: Moving average.

2026-08-19: spike

Output: 1,773.36 tonnes. Tests: IQR, z-score, Moving average, Grubbs.

2026-08-20: spike

Output: 1,965.61 tonnes. Tests: IQR, z-score, Moving average, Grubbs.

2026-08-21: spike

Output: 1,685.59 tonnes. Tests: IQR, Moving average, Grubbs.

2026-08-23: drop

Output: 671.73 tonnes. Tests: Moving average.

2026-08-30: drop

Output: 571.80 tonnes. Tests: Moving average.

2026-09-06: drop

Output: 506.15 tonnes. Tests: Moving average.

2026-09-07: drop

Output: 540.88 tonnes. Tests: Moving average.

2026-09-08: drop

Output: 347.55 tonnes. Tests: IQR, Moving average.

2026-09-11: spike

Output: 1,050.31 tonnes. Tests: Moving average.

2026-09-14: spike

Output: 1,090.95 tonnes. Tests: Moving average.

2026-09-15: spike

Output: 1,084.00 tonnes. Tests: Moving average.

2026-10-04: drop

Output: 582.45 tonnes. Tests: Moving average.

2026-10-11: drop

Output: 575.88 tonnes. Tests: Moving average.

2026-10-17: drop

Output: 223.25 tonnes. Tests: IQR, Moving average.

2026-10-18: drop

Output: 430.95 tonnes. Tests: Moving average.

2026-10-22: spike

Output: 1,165.52 tonnes. Tests: Moving average.

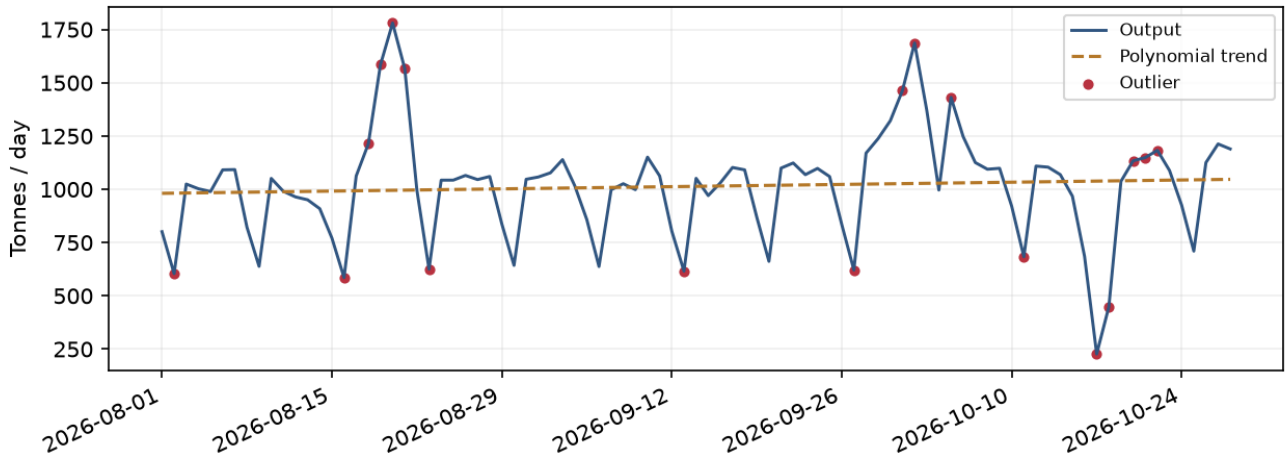
2026-10-23: spike

Output: 1,103.59 tonnes. Tests: Moving average.

Previous N observed rows; current point excluded; no test before a full window is available.
Iterative two-sided Grubbs test; assumes an approximately normal baseline.

Mine 03

Metric	Value
Observations	89
Total	90,233.40
Mean	1,013.86
Standard Deviation	252.96
Median	1,047.00
Iqr	200.06



Detected anomalous observations: 18

2026-08-02: drop

Output: 604.25 tonnes. Tests: IQR.

2026-08-16: drop

Output: 585.46 tonnes. Tests: IQR.

2026-08-18: spike

Output: 1,213.02 tonnes. Tests: Moving average.

2026-08-19: spike

Output: 1,585.09 tonnes. Tests: IQR, Moving average.

2026-08-20: spike

Output: 1,780.55 tonnes. Tests: IQR, z-score, Moving average.

2026-08-21: spike

Output: 1,566.19 tonnes. Tests: IQR, Moving average.

2026-08-23: drop

Output: 625.00 tonnes. Tests: Moving average.

2026-09-13: drop

Output: 615.92 tonnes. Tests: Moving average.

2026-09-27: drop

Output: 620.97 tonnes. Tests: Moving average.

2026-10-01: spike

Output: 1,463.60 tonnes. Tests: IQR, Moving average.

2026-10-02: spike

Output: 1,686.70 tonnes. Tests: IQR, Moving average.

2026-10-05: spike

Output: 1,431.28 tonnes. Tests: IQR.

2026-10-11: drop

Output: 683.22 tonnes. Tests: Moving average.

2026-10-17: drop

Output: 226.92 tonnes. Tests: IQR, z-score, Moving average.

2026-10-18: drop

Output: 447.94 tonnes. Tests: IQR, Moving average.

2026-10-20: spike

Output: 1,133.55 tonnes. Tests: Moving average.

2026-10-21: spike

Output: 1,148.78 tonnes. Tests: Moving average.

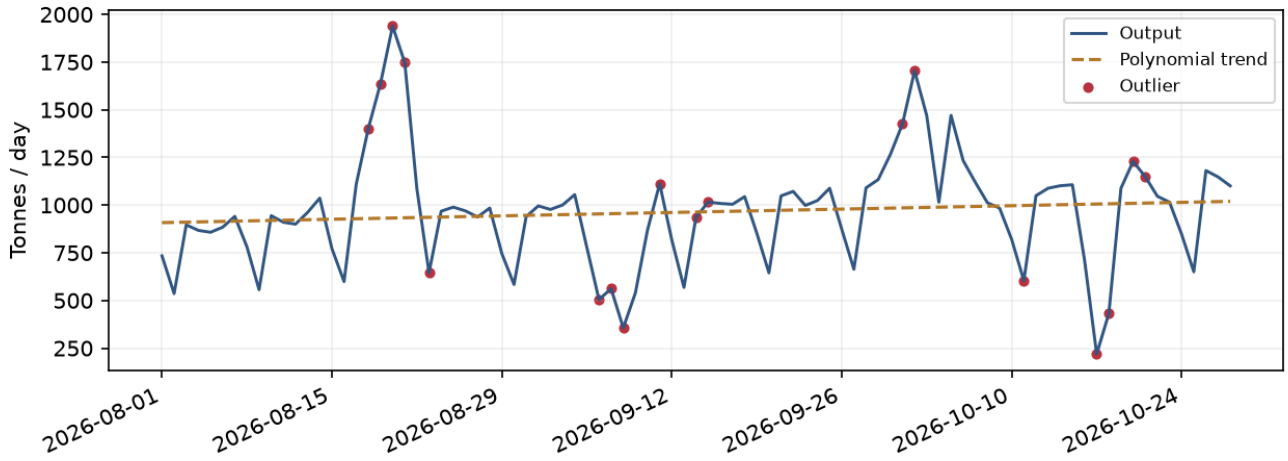
2026-10-22: spike

Output: 1,182.93 tonnes. Tests: Moving average.

Previous N observed rows; current point excluded; no test before a full window is available.
Iterative two-sided Grubbs test; assumes an approximately normal baseline.

Mine 04

Metric	Value
Observations	89
Total	85,789.73
Mean	963.93
Standard Deviation	292.16
Median	984.66
Iqr	271.50



Detected anomalous observations: 18

2026-08-18: spike

Output: 1,402.31 tonnes. Tests: Moving average.

2026-08-19: spike

Output: 1,637.23 tonnes. Tests: IQR, Moving average.

2026-08-20: spike

Output: 1,938.45 tonnes. Tests: IQR, z-score, Moving average.

2026-08-21: spike

Output: 1,746.96 tonnes. Tests: IQR, Moving average.

2026-08-23: drop

Output: 645.34 tonnes. Tests: Moving average.

2026-09-06: drop

Output: 504.47 tonnes. Tests: Moving average.

2026-09-07: drop

Output: 562.10 tonnes. Tests: Moving average.

2026-09-08: drop

Output: 356.32 tonnes. Tests: IQR, Moving average.

2026-09-11: spike

Output: 1,112.37 tonnes. Tests: Moving average.

2026-09-14: spike

Output: 937.54 tonnes. Tests: Moving average.

2026-09-15: spike

Output: 1,015.99 tonnes. Tests: Moving average.

2026-10-01: spike

Output: 1,425.42 tonnes. Tests: Moving average.

2026-10-02: spike

Output: 1,703.92 tonnes. Tests: IQR, Moving average.

2026-10-11: drop

Output: 603.88 tonnes. Tests: Moving average.

2026-10-17: drop

Output: 218.68 tonnes. Tests: IQR, Moving average.

2026-10-18: drop

Output: 431.63 tonnes. Tests: Moving average.

2026-10-20: spike

Output: 1,228.80 tonnes. Tests: Moving average.

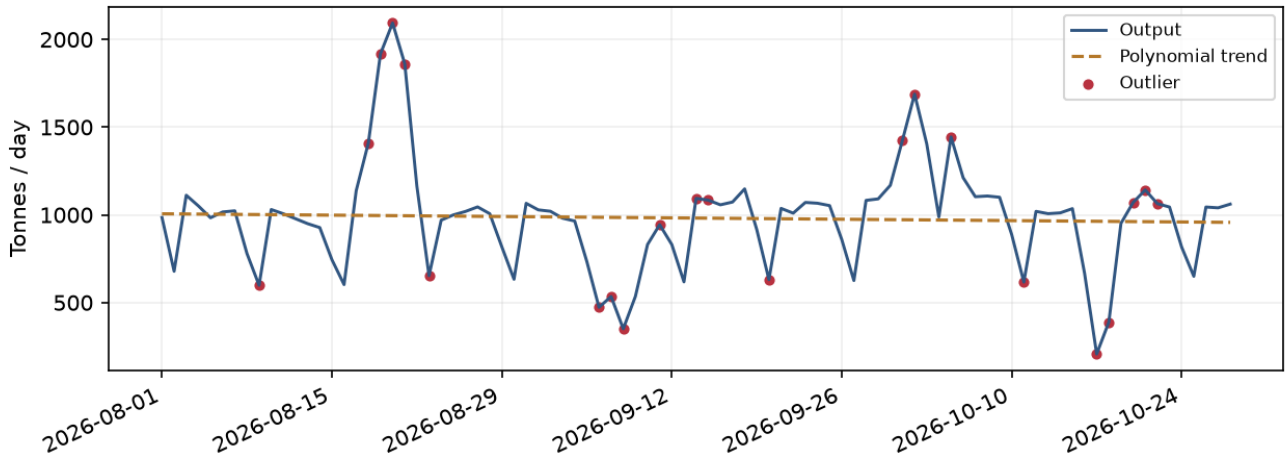
2026-10-21: spike

Output: 1,149.37 tonnes. Tests: Moving average.

Previous N observed rows; current point excluded; no test before a full window is available.
Iterative two-sided Grubbs test; assumes an approximately normal baseline.

Mine 05

Metric	Value
Observations	89
Total	87,364.98
Mean	981.63
Standard Deviation	303.65
Median	1,016.42
Iqr	242.40



Detected anomalous observations: 22

2026-08-09: drop

Output: 598.53 tonnes. Tests: Moving average.

2026-08-18: spike

Output: 1,409.25 tonnes. Tests: Moving average.

2026-08-19: spike

Output: 1,918.08 tonnes. Tests: IQR, z-score, Moving average, Grubbs.

2026-08-20: spike

Output: 2,093.65 tonnes. Tests: IQR, z-score, Moving average, Grubbs.

2026-08-21: spike

Output: 1,860.91 tonnes. Tests: IQR, Moving average, Grubbs.

2026-08-23: drop

Output: 653.14 tonnes. Tests: Moving average.

2026-09-06: drop

Output: 473.26 tonnes. Tests: Moving average.

2026-09-07: drop

Output: 534.61 tonnes. Tests: Moving average.

2026-09-08: drop

Output: 350.49 tonnes. Tests: IQR, Moving average.

2026-09-11: spike

Output: 943.88 tonnes. Tests: Moving average.

2026-09-14: spike

Output: 1,093.90 tonnes. Tests: Moving average.

2026-09-15: spike

Output: 1,083.93 tonnes. Tests: Moving average.

2026-09-20: drop

Output: 630.83 tonnes. Tests: Moving average.

2026-10-01: spike

Output: 1,425.38 tonnes. Tests: Moving average.

2026-10-02: spike

Output: 1,688.27 tonnes. Tests: IQR, Moving average.

2026-10-05: spike

Output: 1,444.07 tonnes. Tests: IQR.

2026-10-11: drop

Output: 616.65 tonnes. Tests: Moving average.

2026-10-17: drop

Output: 206.81 tonnes. Tests: IQR, Moving average.

2026-10-18: drop

Output: 388.56 tonnes. Tests: IQR, Moving average.

2026-10-20: spike

Output: 1,070.33 tonnes. Tests: Moving average.

2026-10-21: spike

Output: 1,141.98 tonnes. Tests: Moving average.

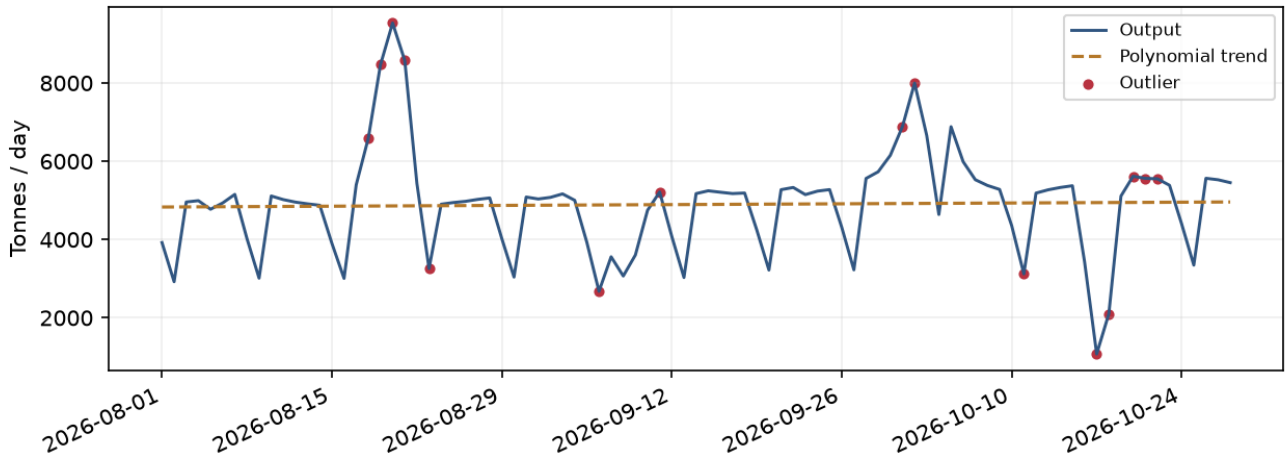
2026-10-22: spike

Output: 1,063.76 tonnes. Tests: Moving average.

Previous N observed rows; current point excluded; no test before a full window is available.
Iterative two-sided Grubbs test; assumes an approximately normal baseline.

Total

Metric	Value
Observations	89
Total	435,631.89
Mean	4,894.74
Standard Deviation	1,324.59
Median	5,076.62
Iqr	1,303.79



Detected anomalous observations: 15

2026-08-18: spike

Output: 6,590.56 tonnes. Tests: Moving average.

2026-08-19: spike

Output: 8,467.35 tonnes. Tests: IQR, Moving average.

2026-08-20: spike

Output: 9,532.86 tonnes. Tests: IQR, z-score, Moving average, Grubbs.

2026-08-21: spike

Output: 8,573.57 tonnes. Tests: IQR, Moving average.

2026-08-23: drop

Output: 3,256.86 tonnes. Tests: Moving average.

2026-09-06: drop

Output: 2,676.26 tonnes. Tests: Moving average.

2026-09-11: spike

Output: 5,215.34 tonnes. Tests: Moving average.

2026-10-01: spike

Output: 6,887.09 tonnes. Tests: Moving average.

2026-10-02: spike

Output: 7,996.30 tonnes. Tests: IQR, Moving average.

2026-10-11: drop

Output: 3,120.94 tonnes. Tests: Moving average.

2026-10-17: drop

Output: 1,074.03 tonnes. Tests: IQR, Moving average.

2026-10-18: drop

Output: 2,101.35 tonnes. Tests: IQR, Moving average.

2026-10-20: spike

Output: 5,616.88 tonnes. Tests: Moving average.

2026-10-21: spike

Output: 5,562.45 tonnes. Tests: Moving average.

2026-10-22: spike

Output: 5,551.57 tonnes. Tests: Moving average.

Previous N observed rows; current point excluded; no test before a full window is available.
Iterative two-sided Grubbs test; assumes an approximately normal baseline.